

ADVITH KRISHNAN

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EDUCATION

B.Tech in Artificial Intelligence

SRM Institute of Science and Technology

Aug 2021 - Aug 2025

CGPA - 8.44/10.00

Relevant Coursework: Neural Networks and Machine Learning, Inferential Statistics, Deep Learning Techniques, Design in Artificial Intelligence Products, Advanced Calculus, Image and Video Processing, Reinforcement Learning.

PROFESSIONAL EXPERIENCE

Systems Engineer (Data & Analytics)

Infosys

Dec 2025 - Present

Chennai, India (Onsite)

- Worked with a Nasdaq-100 & Fortune-500 logistics company on large-scale data pipelines and backend systems.
- Facilitated legacy EDW migration (Informatica, Teradata), handling ~9 billion rows, to Microsoft Azure.

Software Engineer

StejasSYS

Apr 2025 - Sept 2025

Atlanta, USA (Remote)

- Deployed AI models using GCP and built CRUD flows connected across Supabase DB and Databricks for data storage, Docker for containerization and Directus for headless CMS, reducing friction for client products.
- Implemented **proprietary E2E RAG, Agentic and Computer Vision features** for clients in the Health industry.
- Designed complementing NextJS frontends and well-documented FastAPI interfaces, adhering to project requirements.
- Owned full-stack delivery for frontend, backend, and DB tasks, apart from primary responsibilities.

Research Intern

ETH Zurich

Aug 2024 - Mar 2025

Zurich, Switzerland (Remote)

- Simulated motor neuron behavior to understand PNS recruitment patterns for a research project under the Institute of Neuroinformatics & [Neural Bionics Lab](#) at Chalmers Institute of Technology using NEURON and NEST frameworks.
- Single-handedly **programmed customizable nerve fiber models** to simulate 3 invasive neurological experiments.
- Explored deployment strategies of modulatory intraneural models on a mixed neuromorphic chip.

AI Engineer Intern

MIOT International

Dec 2023 - Jul 2024

Chennai, India (Onsite)

- Researched and implemented **2 anchor few-shot models using PyTorch, Weights and Biases, etc.** for shoulder-bone loss estimation & identification of shoulder dislocation type, utilizing only 60 patient scans for training.
- Fine-tuned models to improve diagnostic consensus across 57 surgeons, **reducing decision-making conflicts**.
- **Created analytics reports with Tableau** and conducted lectures on deep learning concepts, making it easier for doctors without a software background to understand the methodologies designed.

Research Intern

NASA

Sep 2022 - Oct 2023

District of Columbia, USA (Remote)

- Performed regression and exploratory data analysis on satellite datasets and applied statistical inference for validation.
- Redesigned and wrangled the **IMPACT team's satellite image data** containing 4 years of sparsely collected top-down multispectral footage of North America by designing a 3-stage data pipeline with arcGIS, Rasterio, Keras, etc.
- Experimented with ResNet34, ResNet50, ResNet104 and Inceptionv3 to validate my data pipeline's effectiveness.

ACHIEVEMENTS

The Linux Foundation | LFX Fellowship Program 2025 | [Proof of Selection](#)

Jun 2025 - Oct 2025

- Selected for the Linux Foundation Fellowship Program 2025 to contribute to the Open Mainframe Project under the Modernization Working Group with a \$3000 stipend (**~1% acceptance rate**).
- Built a RAG framework with data connectors to IBM Z mainframes, IBM IMS mainframes, and modern databases, enabling unified retrieval for Root Cause Analysis and trend analysis on **large-scale infrastructure systems**. Published findings at the Open Mainframe Project (Oct 2025).

PUBLICATIONS

[1] **A. Krishnan**, V. R. Viswasadhas, and Open Mainframe Project, "RAG to Riches: Understanding Actionable Insights from Mainframe Data;," Oct. 2025. — [Publication URL](#)

[2] Pranav Gupta, **Advith Krishnan**, et al. 2025. ViDAS: Vision-based Danger Assessment and Scoring. In Proceedings of the Fifteenth Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2024). Association for Computing Machinery, New York, NY, USA, Article 29, 1–9. — [Publication URL](#)

[3] A. L. Amutha, **A. Krishnan**, et al. "Optimizing smoke detection: A hybrid U-Net Approach with advanced CNN models through satellite image segmentation" (*Under review by TFSC Journal*)

[4] **Advith Krishnan** and Saad Yunus Sait. "Understanding Fine-grained classification with Deep Learning through Fish Species Identification" (*Bachelor Thesis*)

PROJECTS

Deep Divergence Graph Kernel | TensorFlow, NetworkX, NumPy, SciPy | [Project Link](#) | [Reference Paper Link](#)
A refactored implementation of the Deep Divergence Graph Kernel algorithm from Google Research

Contrastive Learner | TensorFlow, PyTorch, NumPy, Keras | [Project Link](#) | [Reference Paper Link](#)
My implementation of the Contrastive Learning-based DenseNet model for Chest X-ray images.

NeuStyle | TensorFlow, PyTorch, NumPy, Keras | [Project Link](#)
An implementation of a Neural Style Transfer model, combined with a VGG-16 Preprocessing backbone.

TouchGrass | Convex, React, TypeScript, TanStack Query, Langchain | [Project Link](#)
An LLM-powered web application for habit-building and personal development.

SKILLS

AI/ML	TensorFlow, LangChain, wandb, OpenCV, Keras, PyTorch, Pinecone, CrewAI Autogen, MLFlow, Ollama, Llama, NumPy, Matplotlib, SciPy, ONNX, Claude, GPT.
Full Stack Development	Databricks, Snowflake, PostgreSQL, Next.js, React, Tanstack Query, REST API, Microservices, FastAPI, Postman, Supabase DB, Datadog, HTTP.
Deployment	Vercel, Amazon Web Services, Google Cloud Platform, Linux, Git, Docker, Kubernetes.
Programming Languages	Python, C, C++, JavaScript/Node.js, TypeScript, SQL, HTML, CSS, Java, Bash.

COMMUNITY/VOLUNTEERING

Board Member Next Tech Lab	Dec 2023 - Aug 2025 <i>Chennai, India (Onsite)</i>
• Drove strategic long-term initiatives for this esteemed and internationally recognized research lab, honored with the prestigious QS (Quacquarelli Symonds) Silver Award for Reimagining Education. • Mentored and lectured 30+ members on EEG/MEG source localization, medical image segmentation, large language models, and spiking neural networks; conducted research paper discussions, hackathons, & talks.	
Research Team Head & Founding Member Odyssey Lab	Dec 2023 - Mar 2025 <i>Chennai, India (Onsite)</i>
• Combined xAI (GradCAM and Integrated Gradients) methods to evaluate generative multimodal LLMs and VLMs (GPT 3.5 Turbo, GPT 4 Turbo, GPT 4o & Gemini 1.5 Pro) in perceiving dangerous scenarios. • Improvised 3 Visual Question Answering (VQA) & Temporal Action Localization pipelines (TAL) to indicate danger in CCTV footage through effective prompt engineering. • Collaborated alongside 9 established vision-language researchers and led 6 interns; presented work at ICVGIP.	